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Maastricht University

Faculty of Sciences
School for Information Technology
Master of Statistics and Data Science

Master's thesis

Developing an in silico transmission model of Vaccinopolis, a clinical research facility, to support the design and conduct of human infection trials

Thanh Chau Pham

Thesis presented in fulfillment of the requirements for the degree of Master of Statistics and Data Science, specialization Biostatistics

SUPERVISOR :

Prof. dr. Niel HENS

Prof. dr. Andrea TORNERI

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Chau, June 2026

Abstract

Objective: Within a small, closed population of 28 individuals (24 volunteers, 4 staff) in Vaccinopolis - a clinical research facility of the University of Antwerp designed for human infection trials - this thesis develops a stochastic individual-based SEIR model to map interpersonal contact patterns, individual viral progression, immunological process, and simulate potential infectious disease transmission.

Methods: The model integrates individual-level heterogeneity in infectiousness, susceptibility, and contact behavior, as well as time-varying contact rates reflecting six daily activity periods to simulate epidemic transmission in continuous time. Four mixing scenarios are used: Free mixing, Staff-bridge, Strict-cohorting, and Common staff room, each is defined by a scenario-specific contact mixing matrix to evaluate daily infection incidence and attack rate distributions across all scenarios. Global sensitivity analysis is performed using Latin Hypercube Sampling (7 parameters, 500 parameter combinations), with Random forest regression and Partial rank correlation coefficients used to determine the most influential parameters. To characterize marginal effects, partial dependence plots are constructed. The impact of four vaccine efficacy levels (30%, 50%, 70% and 90%) at 50% coverage among volunteers is assessed in the placebo-controlled vaccination analysis.

Results: Across 1,000 simulations per scenario, mixing structure substantially influences severity of outbreak dynamics: Scenario 1 Free mixing has the highest attack rates, while Scenario 3 Strict-cohorting limits outbreaks to individual cohorts (rarely exceeding 50%). Scenario 2 Staff-bridge shows that shared staff can bridge isolated groups, while even a 10% inter-staff contact probability in Scenario 4 in the common staff room is sufficient to produce secondary cross-cohort transmission channels. Sensitivity analysis consistently identifies the basic reproduction number R_0 and the infectiousness heterogeneity parameter k as the two dominant drivers of attack rate variability. Vaccination at 50% coverage among volunteers reduces attack rates in all scenarios, with the most pronounced reduction observed in attack rate distributions as vaccine efficacy increases in Scenario 1 Free mixing.

Conclusions: Due to the high stochasticity in the small population such as Vaccinopolis, the contact structure has limited influence on the onset of the outbreak but clearly determines its severity once established. Scenario 3 Strict-cohorting provides the most effective structural containment, while vaccination offers complementary protection. These results can inform the design of quarantine protocols and vaccination strategies for human infection trials at Vaccinopolis.

Keywords: Vaccinopolis, individual-based SEIR model, attack rates, contact structure, sensitivity analysis, vaccination

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Terms and Abbreviations

Terms	**Abbreviations**
AR	Attack rate
CI	Confidence interval
%IncMSE	Percentage increase in mean squared error
IQR	Interquartile range
LHS	Latin Hypercube Sampling
OOB	Out-of-bag
PDF	Probability density function
PDP	Partial dependence plots
PRCC	Partial rank correlation coefficients
R_0	Basic reproduction number
RF	Random forest
SEIR	Susceptible–exposed–infectious–recovered
SIR	Susceptible–infectious–recovered
VE	Vaccine efficacy

1 Introduction

Infectious diseases remain a challenge for global public health, especially after the COVID-19 pandemic, requiring epidemiologists to develop strategies to control transmission dynamics. Mathematical and computational models play an important role in epidemiology, providing a framework to understand the spread of pathogens and to evaluate possible control measures. The benefit of in silico disease modelling is that these models not only provide insight into how a complex pathogen spread process works — it is difficult to understand each component of this process individually —but also predict potential outcomes over time or when specific components of the system change [1].

In recent years, stochastic individual-based models have drawn more interest because they enable the explicit representation of heterogeneity in contacts and disease progression among individuals [2, 3]. For respiratory pathogens, individual variation in infectiousness is particularly important: empirical evidence from SARS-CoV-2 demonstrates that a small proportion of infected individuals generate a disproportionately large number of secondary cases (superspreading events), a pattern characterised by low values of the dispersion parameter k [4, 5, 6]. Transmission of infectious diseases depends on the number of infectious, susceptible individuals in the population and their effective interpersonal contacts [2, 3]

To investigate pathogen transmission, immune responses, and vaccine efficacy under well-defined experimental conditions, controlled human infection studies are conducted in specialized research facilities designed to safely monitor participants throughout the study. Vaccinopolis is an innovative vaccine research center, providing high-quality translational clinical studies aimed at accelerating and sustaining the prevention and control of infectious diseases [7]. Although these settings offer opportunities to investigate the dynamics of infection, they also require careful assessment of transmission risks between individuals within the institution to prevent outbreaks.

The aim of this thesis is to develop an in silico transmission model tailored to the Vaccinopolis quarantine unit. The model integrates three key components: a contact process with time-varying contact rates reflecting the daily operational schedule of the facility, individual-level heterogeneity in infectiousness, susceptibility, and contact behaviour, and structured mixing patterns representing different spatial configurations of volunteers and staff. Beginning with a simplified representation of person-to-person interactions, the model will then be expanded by integrating behavioral aspects and daily routines of individuals within the facility.Four mixing scenarios of increasing containment stringency are evaluated, ranging from unrestricted free mixing to strict cohorting with dedicated staff, to quantify the relationship between contact structure and outbreak risk. A global sensitivity analysis using Latin Hypercube Sampling, Random Forest regression, and Partial Rank Correlation Coefficients is performed to identify the most influential parameters driving outbreak dynamics. Finally, a placebo-controlled vaccination analysis evaluates the impact of partial

vaccination at four efficacy levels on outbreak probability and severity across all mixing scenarios. By linking these simulations at Vaccinopolis, the proposed model aims to generate deeper insights into transmission dynamics and to support the design and evaluation of human infection trials in this unique research setting.

2 Methodology

2.1 Study setting: Vaccinopolis

Vaccinopolis is a vaccine research center located at the University of Antwerp, Belgium, where it conducts two types of clinical studies: ambulatory studies, in which subjects receive a vaccine and return home after a short observation period; and human infection trials, volunteers are exposed to a controlled dose of virus under clinical supervision to monitor viral replication and immune responses [7].

In this study, the simulated population consists of 28 individuals, comprising 24 volunteers and 4 staff members. Volunteers are admitted to a dedicated quarantine unit within the facility for one to three weeks and can possibly move freely within the quarantine floor, including shared dining areas and communal recreation spaces [8]. This design represents a realistic trial size within the capacity of the Vaccinopolis quarantine unit (maximum 30 rooms), reflecting practical enrollment constraints typical of human challenge trial protocols where logistical, ethical, and safety considerations limit participant numbers.

2.2 Model framework

In Figure 1, the SEIR model contains four disease states: susceptible (S), exposed (E), infected (I), and recovered (R) [9, 10].
- Susceptible (S): individuals who are not infected but are at risk of infection.
- Exposed (E): during the latent period drawn from a Weibull distribution ($\text{mean}_e = 5.2$ days, SD = 2.8 days), the viral load is replicating but remains below the threshold required for shedding, so individuals infected but not yet infectious. In this simulation model, we assume that the latent period and the incubation period coincide.
- Infectious (I): individuals who are infected and capable of transmitting the pathogen.
- Recovered (R): individuals who have recovered from infection after a fixed 18-day symptomatic period.

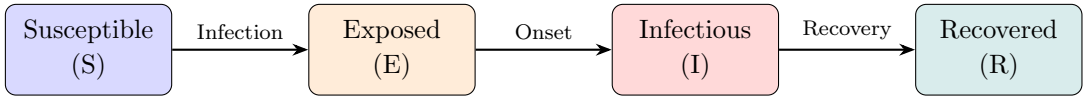


Figure 1: SEIR model state transitions

A stochastic individual-based transmission model is used in this study to simulate the spread of an infectious disease within a closed population. The model represents individuals explicitly and tracks infection events over time, allowing investigation of transmission dynamics arising from interpersonal contacts and time-varying infectiousness profiles.

2.3 Contact process

Contacts between individuals are described by the model in continuous time. More precisely, each infectious individual i generates contacts according to a Poisson process with rate $\lambda_i(t)$, which represents the instantaneous contact rate of that individual at time t. Under this formulation, the waiting time between two consecutive contacts follows an exponential distribution

$$\Delta t \sim \text{Exponential}(\lambda_i(t)).$$

At each contact event, another individual j is selected as the contact partner, either uniformly at random from the remaining population (Scenario 1) or according to a scenario-specific mixing matrix (Scenario 2 to 4, Section 2.4).

To capture the real dynamics of daily life within the facility, the model integrates heterogeneity of exposure over time by applying time-specific multipliers that account for varying levels of activity throughout 24 hours. The day is divided into six distinct functional blocks, each assigned a contact multiplier (M) based on the intensity of social interactions and work tasks for two primary roles: Staff and Volunteers. Staff members are facility personnel responsible for clinical monitoring, sample collection, meal service, and daily operations, who enter the quarantine floor during working hours and interact with volunteers according to a structured daily schedule. Staff members are more likely to get infected during working hours (morning/afternoon) [11], while volunteers are more likely to get infected during group activities. The distribution of these multipliers is summarized in Table 1. The multiplier values in this table are based on the assumption of the typical daily activities and empirical evidence from hospital contact study [11], rather than direct contact measurements in Vaccinopolis.

Table 1: Contact multipliers of Staff and Volunteers

Time Block	**Duration**	**Period**	**Staff Multiplier**	**Volunteer Multiplier**
Morning	5h	06:00 – 11:00	1.5	1.2
Lunch	2h	11:00 – 13:00	1.0	2.0
Afternoon	5h	13:00 – 18:00	1.5	1.0
Dinner	2h	18:00 – 20:00	1.0	1.8
Evening	2h	20:00 – 22:00	0.8	1.5
Night	8h	22:00 – 06:00	0.6	0.5

The dynamic transmission rate at time t for an individual is calculated as:

$$\lambda_{role}(t) = \lambda_{0,i} \times M_{role}(block(t))$$

where $\lambda_{0,i}$ is the individual baseline contact rate, and M_{role} is the multiplier corresponding to the individual's role and the current time block. Baseline contact rates are drawn from a Gamma distribution to capture individual variation in social behavior:

$$\lambda_{0,i} \sim \text{Gamma}(k_{ct},\, ct.rates/k_{ct})$$

where k_{ct} controls the degree of heterogeneity across individuals.

The average multiplier for each role ($\bar{M}_{role}$) over a 24-hour period is computed using a weighted mean:

$$\bar{M}_{role} = \frac{\sum_{i=1}^{6}(M_{i,role} \times t_i)}{24}$$

where t_i represents the duration (in hours) of each time block.

The overall population adjustment factor ($\bar{M}_{total}$) is derived by weighting these averages according to the population size of each group ($N_{staff} = 4$ and $N_{vol} = 24$) to ensure that time blocks do not systematically bias the intended basic reproduction number (R_0):

$$\bar{M}_{total} = \frac{(\bar{M}_{staff} \times N_{staff}) + (\bar{M}_{vol} \times N_{vol})}{N_{total}} \approx 1.065$$

The transmission probability per-contact Q is calibrated as:

$$Q = \frac{R_0}{\bar{\lambda} \times \bar{M}_{total}}$$

where $\bar{\lambda}$ is the mean baseline contact rate of the population. This calibration ensures that the realized R_0 in the complex model remains consistent with the theoretical R_0 defined in the baseline scenario.

2.4 Simulation scenarios

2.4.1 Scenario 1 (Free mixing)

In scenario 1, there is no spatial or social structure. Every individual (volunteers or staff) has an equal probability of contacting any other person in the facility

$$P(j \mid i) = \frac{1}{n-1} = \frac{1}{27}$$

for any individual $j \neq i$. This is equivalent to a single-group mixing matrix $M = [1]$, in which every individual has the same probability of being selected. This represents the highest risk (worst-case scenario), as a single infection can spread rapidly across the entire population without any barriers.

2.4.2 Scenario 2 (Staff-bridge)

The model can be extended by stratifying the population into groups to define a structured mixing process. Each value in the mixing matrix is the probability that a contact made by an individual in group h (row) takes place with an individual in group g (column). Rows sum to one.

Table 2: Contact probability matrix for Scenario 2

	Contact probability in group		
	Vol A	**Vol B**	**Staff**
Vol A	0.9	0.0	0.1
Vol B	0.0	0.9	0.1
Staff	0.4	0.4	0.2

Table 2 shows the contact probability matrix for Scenario 2 Staff-bridge. In this scenario, the population is divided into three groups: half of the volunteers (12 volunteers) in Room A, the other half in Room B, and a shared staff group (4 Staff). From the first row, a contact makes by a volunteer in Room A has a 90% probability of being with volunteer in the same group, 0% with a Room B volunteer, and 10% with a staff member. Staff distribute contacts across both rooms equally (40% each) with 20% among other staff, so they can carry the virus by chance from Room A to Room B. This model simulates cross-contamination risk showing that even if volunteers are isolated, shared personnel can still facilitate a facility-wide outbreak.

2.4.3 Scenario 3 (Strict-cohorting)

Table 3: Contact probability matrix for Scenario 3

	Contact probability in group			
	Vol A	**Vol B**	**Staff A**	**Staff B**
Vol A	0.9	0.0	0.1	0.0
Vol B	0.0	0.9	0.0	0.1
Staff A	0.5	0.0	0.5	0.0
Staff B	0.0	0.5	0.0	0.5

The population is split into two completely independent cohorts: Cohort A including 12 Volunteers A and 2 Staff A, while Cohort B including 12 Volunteers B and 2 Staff B. Staff A never enters Room B, and vice versa. Volunteer cohorts in Table 3 are fully isolated from

each other (90% within-room contact), each retaining exclusive contact with their assigned staff member (10%). Staff A allocates 50% of contacts to volunteer A, and the within-cohort staff probability is 50% (and vice versa). This scenario represents optimal safety. If an infection occurs in Cohort A, it is mathematically blocked from reaching Cohort B. This demonstrates the effectiveness of "zoning" and dedicated staffing in high-security units like Vaccinopolis.

2.4.4 Scenario 4 (Common staff room)

Table 4: Contact probability matrix for Scenario 4

	Contact probability in group			
	Vol A	**Vol B**	**Staff A**	**Staff B**
Vol A	0.9	0.0	0.1	0.0
Vol B	0.0	0.9	0.0	0.1
Staff A	0.45	0.0	0.45	0.1
Staff B	0.0	0.45	0.1	0.45

This scenario extends Scenario 3 by introducing one modification: the two staff groups are permitted to interact in a shared common room, with a 10% inter-staff contact probability allocated in each direction. From Table 4, volunteer rows remain unchanged, while staff contact is redistributed accordingly: 45% within their assigned cohort, 45% among their own staff group, and 10% with the opposing staff group.

This inter-staff contact creates a secondary transmission pathway absent in Scenario 3. If an infection reaches Staff A from Cohort A, it may subsequently be carried to Staff B via the common room, who then re-enters Cohort B as a potential source of cross-contamination. Scenario 4 therefore models a real-world operation because complete isolation of employees outside of working hours is rarely enforceable in practice.

2.5 Infectiousness profile

Rather than assuming a constant rate of transmission during the infectious stage, the model captures the time-varying nature of viral shedding by defining a temporal infectiousness profile. To achieve a flexible representation of the initial rise and subsequent decline in shedding capacity, the infectiousness profile $\nu(t)$ is modeled directly as the probability density function (PDF) of a Gamma distribution:

$$\nu(t) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}$$

where t represents the time elapsed since the individual transitioned from the Exposed (E) into the Infectious (I) compartment, thereby establishing the structural link between the duration of the latent period and the onset of active transmission.

This functional shape is assumed to be identical across all individuals at the population level, acting as a standardized baseline trajectory for every patient. Furthermore, to align this continuous formulation with the discrete, finite duration of the infectious period modeled in the simulation, the function is dynamically normalized over the specific length of the infectious period (t_{inf}) for each individual, ensuring that:

$$\int_0^{t_{\text{inf}}} \nu(t)\, dt = 1$$

This formulation effectively restricts the transmission weight to the active window of viral shedding. After this infectious period expires, individual i recovers and transitions to the Recovered state (R). The resulting probability density profile representing this baseline temporal trajectory over 18 days is visually illustrated in Figure 12 in Appendix.

2.6 Transmission process

The baseline transmission parameter q is calibrated such that the expected number of secondary infections in a fully susceptible population equals the basic reproduction number R_0.

- Infectiousness Multiplier: $z_{\text{inf},i} \sim \text{Gamma}(k, \text{scale} = 1/k)$, where $E[z_{\text{inf},i}] = 1$ and $\text{Var}(z_{\text{inf},i}) = 1/k$. This parameter captures overdispersion in onward transmission potential. Small values of the shape parameter k yield a highly skewed offspring distribution, isolating the impact of superspreading events where a minority of individuals drive the majority of secondary cases [5, 6].
- Susceptibility Multiplier: $z_{\text{sus},j} \sim \text{Gamma}(k_{\text{sus}}, \text{scale} = 1/k_{\text{sus}})$, where $E[z_{\text{sus},j}] = 1$ and $\text{Var}(z_{\text{sus},j}) = 1/k_{\text{sus}}$. This parameter accounts for individual variations in biological vulnerability or personal protective barriers against acquiring the pathogen.

If an infectious individual i contacts a susceptible individual j at time t, transmission occurs with probability

$$p_{ij}(t) = q\, z_i\, \nu(t - t_i^{\text{inf}}),$$

where
- q is the baseline transmission parameter,
- z_i is an individual infectiousness multiplier,
- t_i^{inf} is the infection time of individual i,
- $\nu(\cdot)$ is the infectiousness profile describing how infectiousness varies with time since infection.

Under this formulation, transmission is governed by three components: the temporal contact process determining interaction timestamps, the within-host infectiousness profile reflecting physiological shedding dynamics, and the multi-dimensional host heterogeneity parameters $(k, k_{\text{sus}}, k_{\text{ct}})$ controlling individual transmission thresholds. This integrates framework provides a robust mechanistic link between micro-level biological progression within infected individuals and the macro-level stochastic spread of the pathogen across the entire cohort. All key structural parameters are systematically varied during the global sensitivity analysis, as summarized in Table 5.

Table 5: Model parameters

Parameters	**Mean value (SD)**	**Distribution**	**Reference**
Population size n	28	NA	Motivated in text
Basic reproduction number R_0	1.5	NA	Assumption; SA
Latent period	5.2 days (2.8 days)	Weibull	[4]
Recovery period	18 days	Exponential	[4]
Daily contact rate $ct.rates$	3 days (2.12 days)	Gamma	[5, 6, 12]
Infectiousness measure $mean_i$	10 days (3.8 days)	Gamma	[4]
Infectivity heterogeneity k	0.2 (2.236)	Gamma $(k = 0.2, 1/k)$	[5, 6]
Individual infectivity z_{inf}	1 (1.414)	Generated via k	[5, 6]
Susceptibility heterogeneity k_{sus}	1 (1)	Gamma $(k_{\text{sus}} = 1, 1/k_{\text{sus}})$	Assumption
Individual susceptibility z_{sus}	1 (1)	Generated via k_{sus}	Assumption
Contact rates heterogeneity k_{ct}	1 (1)	Gamma $(k_{\text{ct}} = 1, 1/k_{\text{ct}})$	Assumption

2.7 Simulation algorithm

The simulation proceeds 1,000 iterations for each scenario. The epidemic process evolves in continuous time through two types of events: contact events and recovery events. The simulation terminates when no other infectious individuals are present in the population. A maximum of 360 days is set as a computational safeguard, although in practice epidemics in a closed, small population reaches completion or go extinct before this limit.

At each step of the simulation, the next event time is determined as

$$t_{\text{next}} = \min(t_{\text{contact}}, t_{\text{recovery}})$$

If the next event is a contact event, a potential transmission is evaluated according to the probability $p_{ij}(t)$. If the next event is a recovery event, the infected individual transitions to the recovered state.

2.8 Calibration and Sensitivity analysis

Sensitivity analysis is performed to quantify the influence of seven epidemiological input parameters (R_0, k, k_{sus}, k_{ct}, $ct.rates$, $mean_e$, $mean_i$) on three outcome measures: total attack rates (AR_{total}), attack rates of volunteers ($AR_{volunteers}$), attack rates of staff (AR_{staff}). All these outcomes are logit-transformed before analyzing because this transformation maps the percentages from the $(0, 1)$ interval to the real line $(-\infty, +\infty)$ to stabilize variance and improve linearity.

$$\text{logit}(AR) = \log\left(\frac{AR}{1 - AR}\right)$$

The seven parameters varying in the analysis are:

- The the basic reproduction number $R_0 \in [1.5, 4.0]$.
- The infectiousness heterogeneity dispersion parameter $k \in [0.1, 1.0]$.
- The susceptibility heterogeneity parameter $k_{sus} \in [0.5, 1.5]$.
- The contact rate heterogeneity parameter $k_{ct} \in [0.5, 1.5]$.
- The daily contact rate $ct.rates \in [1.0, 6.0]$.
- The mean latent period $mean_e \in [3.0, 7.0]$ days.
- The mean infectious period $mean_i \in [5.0, 12.0]$ days.

2.8.1 Calibration check

In this simulation study, Volunteers and Staff are given different multiplier profiles with time-varying contact rates that differ across six daily activity periods (morning, lunch, afternoon, dinner, evening, and night). To make sure that these temporal modifiers do not systematically bias the effective reproduction number away from the input value R_0, the weighted mean of all time-block multipliers is calculated by weighting the number of hours in each block.

The calibration is verified by comparing the attack rate distribution obtained under two conditions across 1,000 Monte Carlo simulations: a baseline model with no time-block adjustment, and the calibrated model incorporating the full time-block structure. Under both conditions, the basic reproduction number is fixed at $R_0 = 1.5$ and all other parameters are constant. If the attack rate distributions between these conditions are nearly identical, then they confirm that the calibration factor successfully preserves the intended R_0 regardless of the time-block structure.

2.8.2 Latin hypercube sampling

In simple random sampling, data points are selected entirely at random. This leads to two risks:

- Clustering: Many data points are placed close together, causing information redundancy with randomly selection.
- Gaps: There are areas of value that have no samples at all, causing the model to miss important data.

Therefore Latin hypercube sampling (LHS) is developed to solve the problem of "space-filling" in the most efficient way because it ensures that the entire range of each input variable is completely covered without regard to which single variable or combination of variables might dominate the model reponses. [13]

Using the Latin hypercube sampling *lhs* package in R, parameter combinations are generated with 500 samples, then LHS will divide the range of values of each of seven variables into 500 equal segments. In total, the theoretical space in LHS has 500^7 possible hypercube units $[0, 1]$:

$$\mathrm{p} = p_{min} + x + (p_{max} - p_{min})$$

where $x \in [0, 1]$ is the LHS sample value and p_{min} and p_{max} are the lower and upper bounds of the parameter range. LHS generates exactly 500 unique parameter combinations (samples) through the following steps [14]:

- Independent Permutation: A list of 500 values is generated for each of seven parameter ranges. These lists are then independently and randomly permuted.
- Pairing: The first value from the shuffled R_0 list is paired with the first value from the shuffled k list, the first from the k_ct list, and so on. This creates Sample 1 and this process repeats until Sample 500 is formed.

For each of the 500 parameter combinations, the simulation is replicated 20 times and the mean AR is computed prior to logit transformation, in order to reduce stochastic noise in the outcome variable before fitting statistical models.

2.8.3 Random forest

Random forest is an ensemble learning method that grows multiple decorrelated trees on bootstrap samples of the data, then averages their predictions to improve predictive accuracy and reduce overfitting [15]. To explore the relationship between epidemiological input parameters and the attack rates AR, random forest regression models are used in this study to capture nonlinear relationships, interactions between variables, and stochastic variability without requiring explicit model specification.

The coefficient of determination, R^2, as a measure of model fit, computes the proportion of variance in the outcome explained by the model:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

where y_i represents observed values, $\hat{y}_i$ predicted values, and $\bar{y}$ the mean of observed values.

The out-of-bag R^2 (OBB R^2) and mean squared error (OBB MSE) are used to evaluate the predictive performance of the random forest models. OOB R^2 is defined as:

$$R^2 = 1 - \frac{\text{MSE}_{\text{OOB}}}{\text{Var}(y)}$$

where MSE_{OOB} is the mean squared prediction error on OOB observations and $\text{Var}(y)$ is the total variance of the outcome. The OOB Error estimate is used as the internal validation index of Random Forest. Since each tree is built using a bootstrap sample, approximately one-third of the observations are not used in its construction. These OOB samples provide an unbiased estimate of prediction error and do not require a separate test set.

Variable Importance is assessed using permutation importance, which means the percentage increase in mean squared error (%IncMSE), to identify the most influential parameters driving outbreak dynamics. The values of each input parameter are randomly permuted in the OOB data, breaking the association between that predictor variable and the outcome. After that, the model is used to generate new predictions, and the resulting increase in prediction error is recorded. A larger %IncMSE indicates that the model relies more on that variable for accurate predictions to explain variability in attack rates; a value near zero or negative indicates negligible predictive contribution [16].

2.8.4 Partial rank correlation coefficients

Partial rank correlation coefficients (PRCC) is a global sensitivity analysis method for models that measures the monotonic association between a given input parameter and the outcome after removing the linear effects of all other parameters [17, 18, 19, 20]. Prior to analysis, both input parameters and outcomes are rank-transformed to reduce the influence of non-normality and non-linearities.

The PRCC for each parameter X_j and outcome Y is computed as the Pearson correlation between the residuals $R(X_j \mid X_{-j})$ from the rank regression of X_j on all other input parameters X_{-j} and the residuals $R(Y \mid X_{-j})$ from the rank regression of the outcome Y on all other input variables X_{-j}.

$$\text{PRCC}_j = \text{cor}\Big(R(X_j \mid X_{-j}),\ R(Y \mid X_{-j})\Big)$$

PRCC values range from -1 to $+1$, where positive values indicate that increasing the parameter leads to higher outcomes, and inverse relationship in case of negative values, while the values near 0 indicate weak or no monotonic association after controlling for other parameters.

The same 500 LHS samples used for the RF analysis are used for PRCC computation, ensuring full consistency between methods.

2.8.5 Partial dependence plots

Partial dependence plots (PDP) are constructed for the two most influential parameters identified by both RF and PRCC (R_0 and k) to visualize the marginal effect of each parameter on the outcome by averaging over the joint distribution of all other parameters [21].

For a variable X_s, the partial dependence function is defined as:

$$\hat{f}_{PD}(X_s) = \frac{1}{n} \sum_{i=1}^{n} \hat{f}(X_s, X_{C,i})$$

where $\hat{f}$ is the fitted model, $X_{C,i}$ is all other covariates for observation i, n is the number of observations.

A two-dimensional PDP is additionally constructed for the joint effect of R_0 and k on $logit(AR_{total})$ under Scenario 1 Free mixing, to visualize potential interaction effects between them.

2.9 Outcomes

Primary outcome variables of the stochastic SEIR model are extracted from each simulation run to characterize the outbreak dynamics under different contact structures and mixing scenarios.

2.9.1 Daily infection incidence

The daily infection incidence is defined as the number of individuals transitioning from the susceptible (S) to the exposed (E) state on a given simulation day.

Daily incidence is computed as:

$$I_t = \sum_{i=1}^{N} 1(t_i \in [t, t+1))$$

where t_i is the onset time of infectiousness for individual i, $1(\cdot)$ is the indicator function, and N is the population size.

Daily incidence curves are used to compare outbreak trajectories across scenarios and to assess the timing and magnitude of epidemic peaks throughout the 360-day simulation period. Daily infection incidence is aggregated to the weekly level by summing over each 7-day window. The weekly prevalence is calculated as the cumulative number of infected individuals at the last day of each week.

2.9.2 Attack rate distribution

The attack rate (AR) is calculated as the proportion of individuals who were infected in Vaccinopolis at the end the outbreak period for each simulation replicate.

The total attack rate is defined as:

$$AR_{\text{Total}} = \frac{N_{\text{infected}}}{N}$$

where N_{infected} is the cumulative number of infected individuals during the simulation, and N is the total population size.

Subgroup-specific attack rates are also calculated separately for Volunteers and Staff:

$$AR_{\text{Volunteers}} = \frac{N_{\text{infected,Volunteers}}}{N_{\text{Volunteers}}}$$

$$AR_{\text{staff}} = \frac{N_{\text{infected,Staff}}}{N_{\text{Staff}}}$$

Outbreak extinction probability is used to quantify the likelihood that transmission chains terminated without generating a major outbreak. We consider an outbreak is extinct if the total number of infected individuals remain below 10% threshold during the simulation period [22]. In a population of $N = 28$ as Vaccinopolis, a major outbreak is defined as a simulation in which at least 3 individuals are infected in total (including the index case), corresponding to at least 2 secondary infections and approximately 10.7% of the population.

Outbreak extinction probability is estimated as:

$$P(\text{extinction}) = P(AR_{\text{total}} \leq 10\%) = 1 - P(AR_{\text{total}} > 10\%)$$

Outbreak extinction probability assess the ability of different intervention scenarios and contact structures to prevent sustained transmission. Higher extinction probabilities indicate greater epidemic control and reduce risk of large outbreaks.

2.9.3 Placebo-controlled vaccination analysis

To assess the impact of vaccination on outbreak dynamics, four vaccine efficacy (VE) levels are evaluated: 30%, 50%, 70%, and 90%, alongside a placebo group (VE = 0). These levels

are chosen to span the range of observed and regulatory-relevant efficacy thresholds for respiratory pathogen vaccines. The World Health Organization (WHO) have set a minimum acceptable efficacy of 50% for COVID-19 vaccines, with a preferred target of at least 70% [23]. The lower bound of 30% corresponds to the minimum precision criterion recommended by the WHO, which requires that the 95% confidence interval of the trial estimate exclude efficacy below 30% [24]. The upper bound of 90% reflects the highest efficacy observed in phase III trial of the Pfizer–BioNTech BNT162b2 vaccine (95.0%, 95% CI: 90.3–97.6) [25]. The intermediate value of 70% is consistent with the Oxford–AstraZeneca ChAdOx1 vaccine (62.1 - 90.0% depending on dosing regimen) [26]. By evaluating these four levels, the analysis captures the full spectrum from minimally effective to highly effective vaccines.

Vaccination is modelled as a multiplicative reduction in individual susceptibility. For each vaccinated individual i, the susceptibility multiplier $z_{sus,i}$:

$$z_{sus,i}^{vacc} = z_{sus,i} \cdot (1 - \varepsilon_{sus})$$

where $z_{sus,i} \sim \text{Gamma}(k_{sus}, 1/k_{sus})$ captures individual-level heterogeneity in susceptibility and $\varepsilon_{sus} \in \{0, 0.30, 0.50, 0.70, 0.90\}$ denotes vaccine efficacy against susceptibility.

Vaccination coverage is set at 50% and among 24 volunteers, 12 are randomly selected for vaccination in each simulation run. Staff members ($n = 4$) remains unvaccinated in all conditions. For each of the four mixing scenarios, every VE level is compared against the placebo condition across 1000 replications. To ensure a fair comparison, both placebo and vaccine groups within each VE level are initialized with the same random seed for each replication, so that the only source of variation between groups is vaccine efficacy. This paired design eliminates confounding due to differences in index case identity, contact sequence, or individual-level heterogeneity draws. The primary outcome is the distribution of attack rates conditional on major outbreak, examined to characterize outbreak severity among outbreaks that are not prevented by vaccination.

2.10 Software

The data analysis is proceeded using R software version 4.5.3. Statistical significance for PRCC is assessed at the 5% level, defined as the 95% bootstrap confidence interval not crossing zero. All simulation code is available at `https://github.com/tcp7497/Master-Thesis---Chau-`

3 Results

3.1 Simulation results

3.1.1 Daily infection incidence

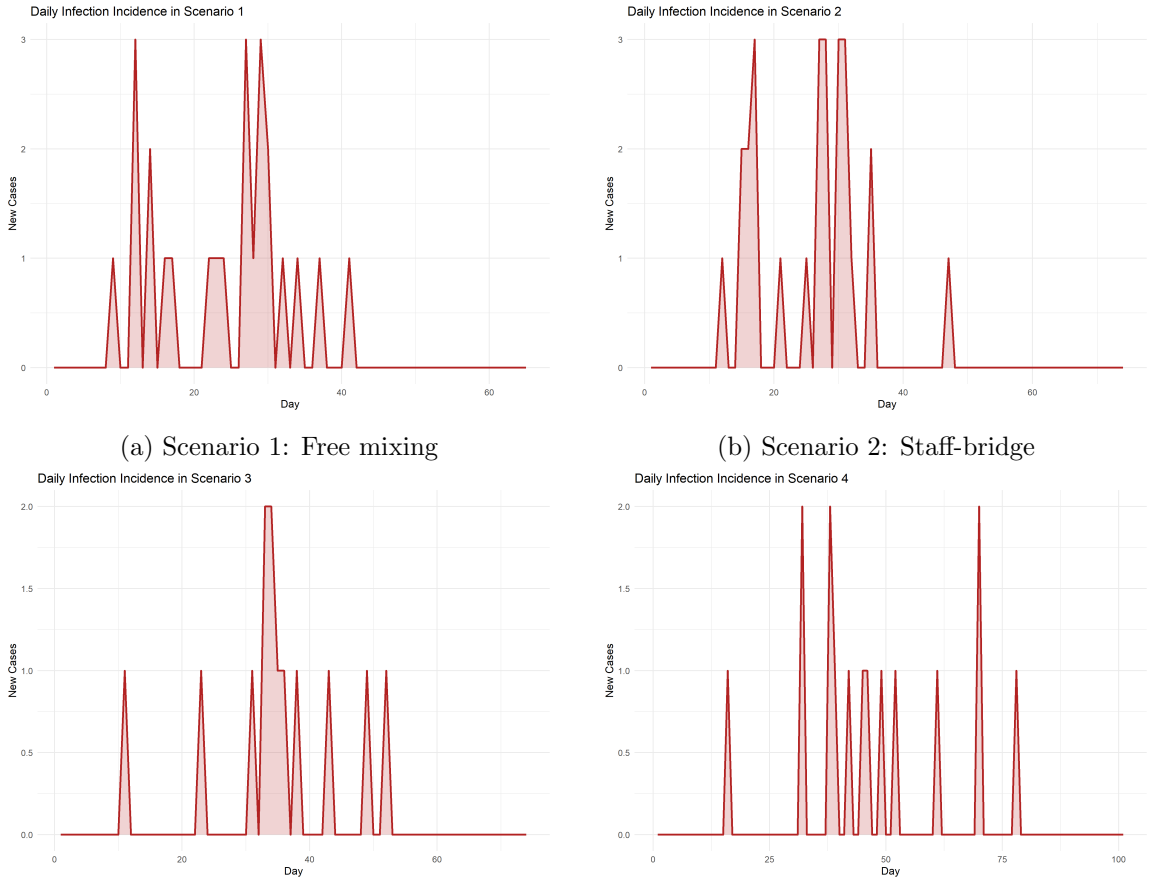
Figure 2 illustrates the daily infection incidence across four mixing scenarios. For each scenario, the simulation run with the highest number of cases among 1000 replicates is displayed to illustrate the epidemic dynamics under maximum outbreak severity.

In Figure 2a Scenario 1 Free mixing, the epidemic curve shows the highest number of new cases between days 10 and 30, peaking at 3 new cases/day for several consecutive days, reflecting rapid, unimpeded transmission through the fully connected population. The absence of any structural barriers allows the virus to quickly infect the susceptible, producing a classic short-duration, high-intensity outbreak.

Figure 2b Scenario 2 Staff-bridge displays new infected cases are more scattered in the early stage with a primary peak followed by a subsequent wave appearing around days 18 – 40 later than in Scenario 1. This delayed secondary peak is the epidemiological evidence of cross-cohort transmission between volunteer groups caused by shared staff: after the initial outbreak in one volunteer room, the virus is carried by a staff member into the second group, starting a new transmission chain in the remaining population with a delay corresponding to the staff incubation and serial interval. Despite the partial isolation of volunteer cohorts, the total number of cases remains substantial, as the staff bridge eventually links both rooms.

The epidemic curve in Figure 2c Scenario 3 Strict-cohorting shows small, low-amplitude peaks distributed over a longer period, with no sustained transmission wave. The strict separation of two groups of volunteers and staff prevents any single infection event from generating a large cluster, and most introductions result in rapid local extinction. When transmission does occur, it is confined entirely within one cohort, with no evidence of cross-cohort spread.

In Figure 2d Scenario 4 Common staff room, the daily infection incidence plot exhibits two distinct outbreaks: the first is strong and occurs early around days 30 – 40, the second appears later around day 70, separated by a short temporal lag. This result reflects the inter-staff transmission pathway introduced by the contact in the common staff room. An infection originating in cohort A reaches staff A, who subsequently transmits to Staff B during a shared break period; staff B then seeds a new cluster in cohort B. In contrast to Scenario 3 where such cross-cohort linkage is structurally impossible, the dual-cohort involvement of Scenario 4 shows that even a modest 10% inter-staff contact probability is sufficient to bridge otherwise isolated cohorts under adverse transmission conditions.



(a) Scenario 1: Free mixing

(b) Scenario 2: Staff-bridge

(c) Scenario 3: Strict-cohorting

(d) Scenario 4: Common staff room

Figure 2: Daily infection incidence in four scenarios

3.1.2 Attack rate distribution

Figure 3 illustrates the conditional attack rate distribution among simulations classified as major outbreaks. Overall, all graphs exhibit a distinctly right-skewed distribution with the frequency peak in four scenarios strongly concentrated around 0.1 – 0.2 proportion infected. Many simulations in Scenario 1 Free mixing reach attack rates of 60–100%, confirming that unrestricted mixing allows the virus to rapidly exhaust the susceptible pool once transmission is established. Moreover, major outbreaks in Scenario 2 Staff-bridge reach attack rates of 60-80% of the population, indicating that cohort splitting reduces the speed and extent of transmission, although the staff bridge still enables facility-wide spread. Besides, the attack rate distribution in Scenario 3 Strict-cohorting is compressed rarely exceeding 50%, as infections remain largely confined to the initially affected cohort. Scenario 4 Common staff room demonstrates a highest frequency peak at 0.1 – 0.2 proportion infected, as well

as an intermediate pattern: the distribution is similar to Scenario 3 at low attack rates, but shows a secondary mass at higher values (40–60%), possibly due to simulations in which contact between staff successfully connected both volunteer groups. This confirms that 10% contact of staff in the common room in Scenario 4 is sufficient to meaningfully elevate the conditional severity of major outbreaks relative to Scenario 3. Total attack rates of four scenarios are also presented in Figure 13 the Appendix.

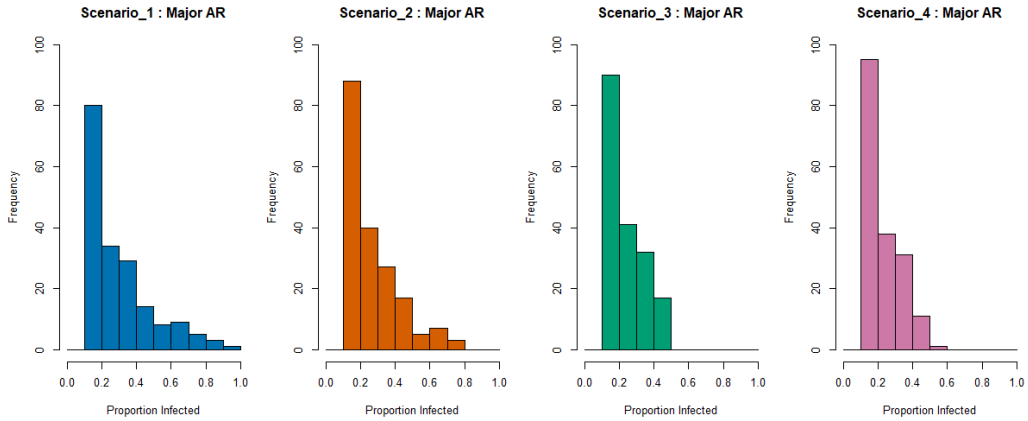


Figure 3: Attack rates of major outbreaks in four scenarios

The extinction probability in Figure 14 in the Appendix is consistently high across all scenarios (more than 80%), suggesting that the mixing structure has limited influence on whether an outbreak occurs, but a substantial influence on what happens once it does.

3.1.3 Vaccination impact

Figure 4 displays the distribution of attack rates conditional on major outbreaks (AR $> 10\%$) in four scenarios, corresponding to five levels of vaccine efficacy (from placebo VE $= 0\%$ to VE $= 90\%$). The plot suggests a downward trend in both the median and the interquartile range (IQR) of attack rates as vaccine efficacy increases and group structures become stricter.

Scenario 1 exhibits the widest spread of attack rates (IQR approximately 0.10 to 0.50, median AR is close to 0.3), reflecting high stochastic variability in the fully connected free-mixing network. As vaccine efficacy increases to 90%, IQRs narrow substantially across all scenarios, with a median AR below 0.20. Under placebo conditions in the free mixing structure, the vaccine is the only biological barrier that prevents disease spread. However, given the small population size of 28 individuals, a clearly overlap between the boxes is shown across several scenarios, reflecting high stochastic uncertainty. For example, in Scenario 2, the trend between boxplots of attack rates in Placebo and VE $= 30\%$ group

does not show visual reduction. Therefore, these findings should be interpreted as suggestive trends rather than statistically definitive effects.

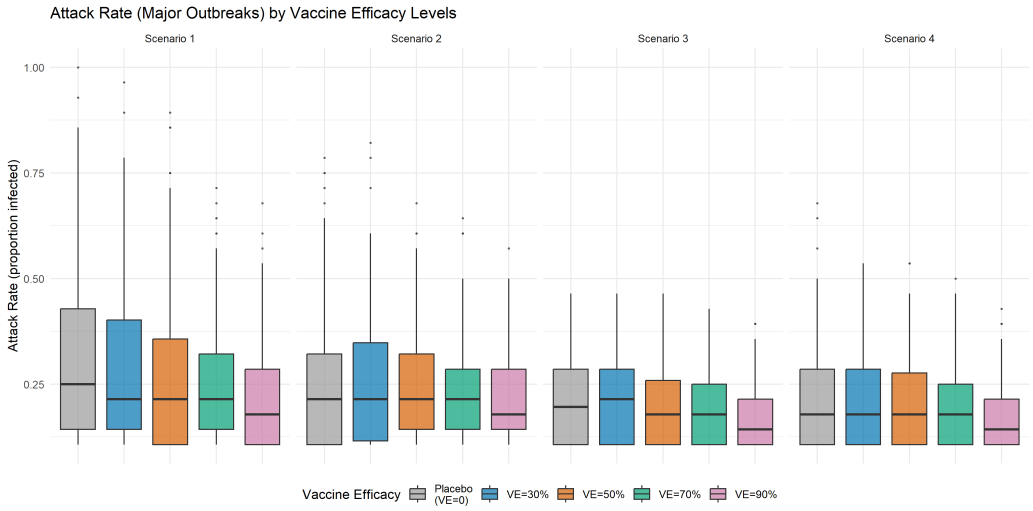


Figure 4: Attack rates of major outbreaks by Vaccine efficacy levels

As detailed in Appendix Table 6, the probability of major outbreak decreases across all scenarios when vaccine efficacy levels increase.

3.2 Calibration and Sensitivity analysis

3.2.1 Calibration

Figure 5 compares the distribution of attack rates conditional on major outbreaks obtained from the model with and without time blocks. Broadly similar distributions are observed in both conditions with nearly identical mean 32% and comparable IQRs of approximately 15–45%. This indicates that the time-block calibration preserves the epidemic dynamics without introducing systematic bias.

However, there is a difference in the violin shape of the distributions: no time blocks model (median AR = 21.4%) with a broader upper tail and thinner base produces slightly more simulations reaching high attack rates above 75% compared to the calibrated model (median AR = 25.0%). This difference is attributable to the overall population adjustment factor $\bar{M}_{total} \approx 1.065 > 1$, which slightly reduces the transmission probability $Q = R_0/(\bar{\lambda} \times \bar{M}_{total})$ relative to the flat baseline no time blocks, thereby moderating the most severe outbreak trajectories in the calibrated model.

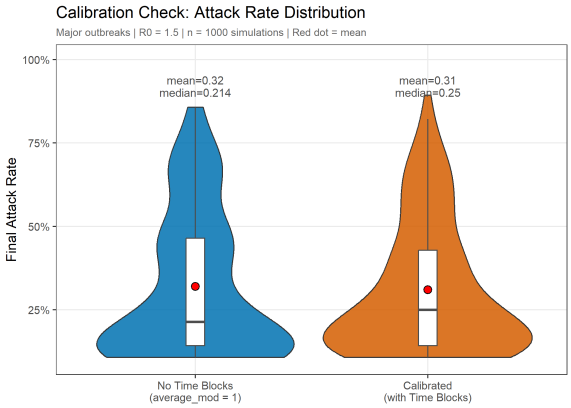


Figure 5: Attack rate distributions between with and without time blocks

3.2.2 Sensitivity analysis

Variable importance

Figure 6 presents the permutation importance (%IncMSE) for all seven input parameters across scenarios and outcomes. As can be seen, the basic reproduction number R_0 is the most influential parameter for all outcomes, with %IncMSE values of 40–65% for AR_{Total} and $AR_{Volunteers}$ and 15–50% for AR_{Staff}. The infectiousness heterogeneity parameter k ranks the second, with %IncMSE values of 35–50% for AR_{Total} and $AR_{Volunteers}$ and 10–40% for AR_{Staff}.

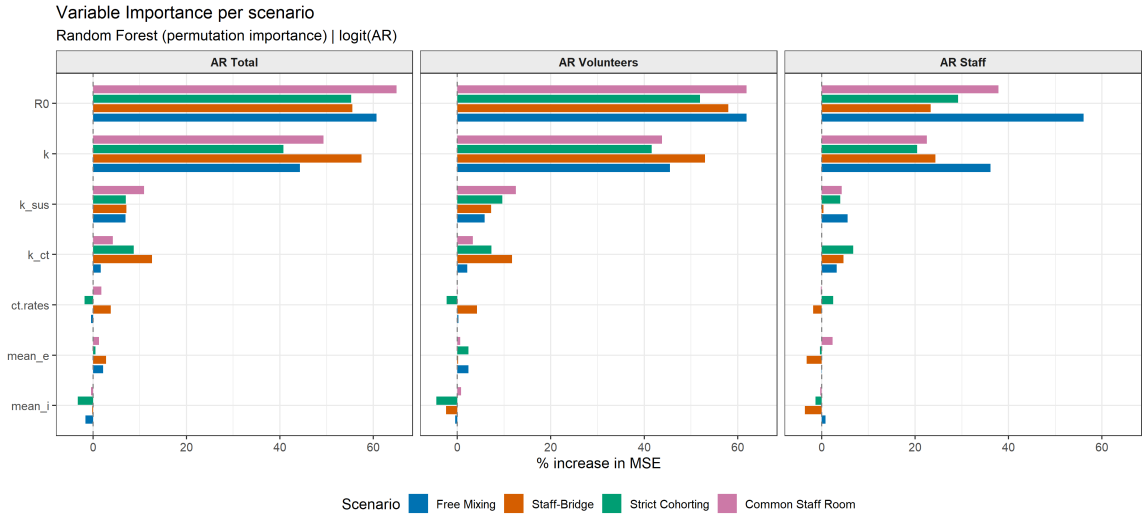


Figure 6: Variable Importance per scenario

Partial rank correlation coefficients

Consistent with the previous result Variable importance, R_0 and k are also the two dominant drivers of attack rate variability across all scenarios in Figure 7. The slight reduction in the influence of R_0 from Scenario 1 Free mixing to Scenario 3 Strict-cohorting reflects in the progressive decrease in PRCC values, suggesting that more restrictive mixing structures partially neutralize the effect of transmissibility by fragmenting the contact network and reducing within-cohort reproduction numbers.

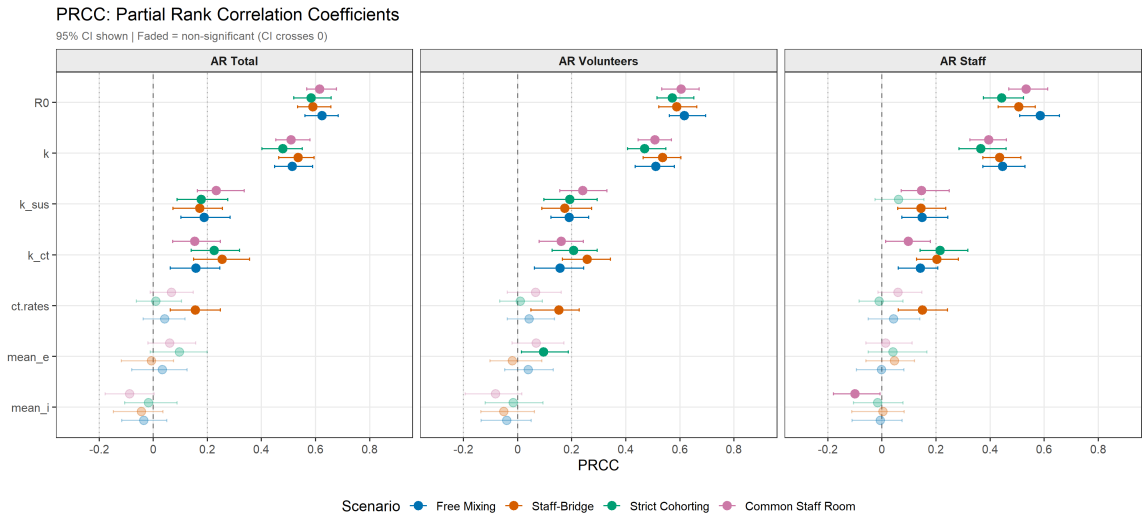


Figure 7: Partial Rank Correlation Coefficients

A supplementary plot of the PRCC values restricted to the two dominant parameters R_0 and k across all scenarios and outcomes is provided in Figure 15 in Appendix. Both parameters maintain positive associations with attack rates, with R_0 achieves higher PRCC values for AR_{Total} and $AR_{Volunteers}$ while k shows slightly larger influence on AR_{Staff} under structured mixing scenarios.

Partial dependence plots

The PDP for R_0 is displayed in the Figure 8. Across all scenarios and outcomes, logit(AR) rises monotonically with R_0, with the steepest increases occurring in the range $R_0 \in [1.5, 2.5]$, followed by a progressive flattening at higher values. This pattern is consistent with the theoretical relationship between R_0 and final epidemic size in a finite population when susceptible depletion limits further spread. Scenario 1 Free Mixing consistently yields the highest logit(AR) values at any given R_0, while Scenario 3 Strict-cohorting produces the lowest, confirming that mixing structure modifies the effective influence of transmissibility.

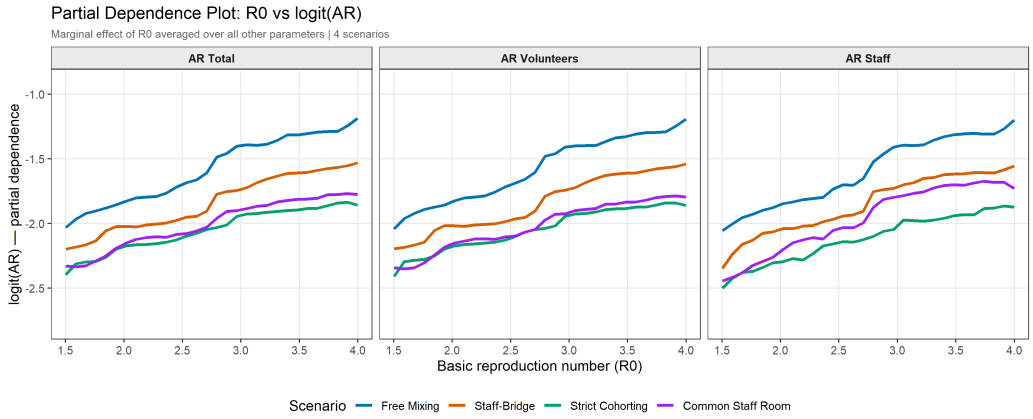


Figure 8: Partial Dependence Plots: logit(AR) vs. R_0

Figure 9 presents the PDP for k. Logit(AR) rises with k across all scenarios, with the steepest increase between 0.1 and 0.5, and goes near-plateau when k exceeds 0.75. This pattern demonstrates that the transition from high to moderate infectiousness heterogeneity has the largest marginal effect on attack rates. The majority of individuals transmit little at very low k (high heterogeneity), and the outbreak depends almost entirely on the rare occurrence of a superspreader event. As k increases toward homogeneity, transmission becomes more reliable and epidemic extinction less likely, yielding higher expected attack rates.

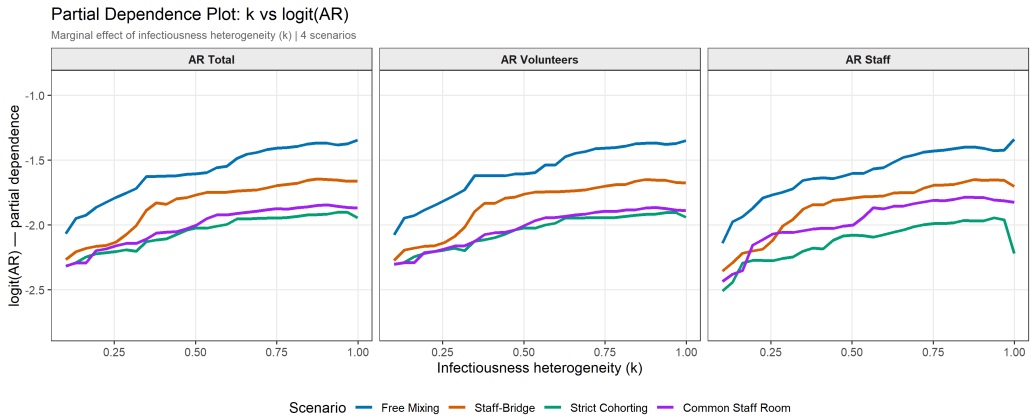


Figure 9: Partial Dependence Plots: logit(AR) vs. k

The two-dimensional PDP (Figure 10) shows the joint non-linear relationship between R_0 and k on the total attack rate under Scenario 1 Free mixing. The contour lines show that the scale of transmission is highly sensitive to changes in both parameters, but their impacts are asymmetric. At low k (< 0.3), the total attack rate remains low, even as R_0 upward

toward 4.0 because superspreading dynamics govern the outbreak outcomes regardless of R_0. However, at higher k (≥ 0.5 more homogeneous), the total attack rate increases rapidly with the highest infection ($\text{logit}(AR) \approx -1.2$) concentrates in the upper-right quadrant.

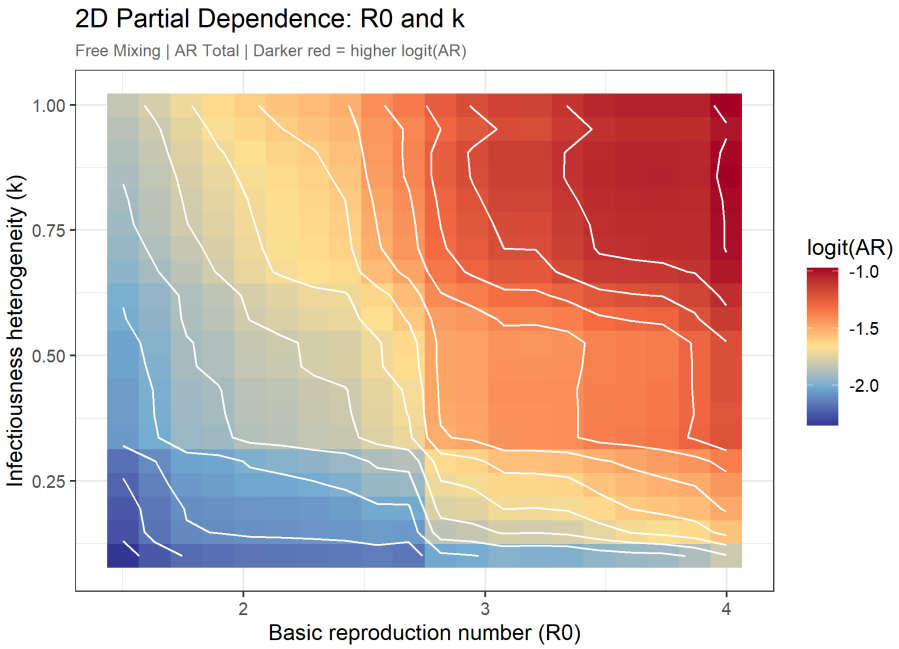


Figure 10: Partial Dependence Plots: R_0 vs. k

Figure 11 shows the strong negative monotonic relationship between the mean baseline contact rate $\bar{\lambda}$ and the transmission probability per-contact Q across the 500 LHS parameter combinations (Spearman $r = -0.863$, $p < 0.001$). From the calibration formula $Q = R_0/(\bar{\lambda} \times \bar{M}_{total})$, increasing the contact rate $\bar{\lambda}$ decreases the transmission probability Q to maintain the intended R_0, and vice versa. The scatter around this trend reflects the variation in R_0 across LHS samples ($R_0 \in [1.5, 4.0]$).

This negative dependence is a consequence of the top-down parameterization method applied in this study, in which R_0 is specified as an input and Q is calibrated accordingly. As a result, $\bar{\lambda}$ and Q are not free to vary independently: their effects on attack rates partially offset each other, which explains the low variable importance of the contact rates parameter in both the Random Forest and PRCC analyzes. This does not indicate that contact frequency is epidemiologically unimportant, but rather that under this parameterization, its effect is absorbed by the compensating adjustment in Q.

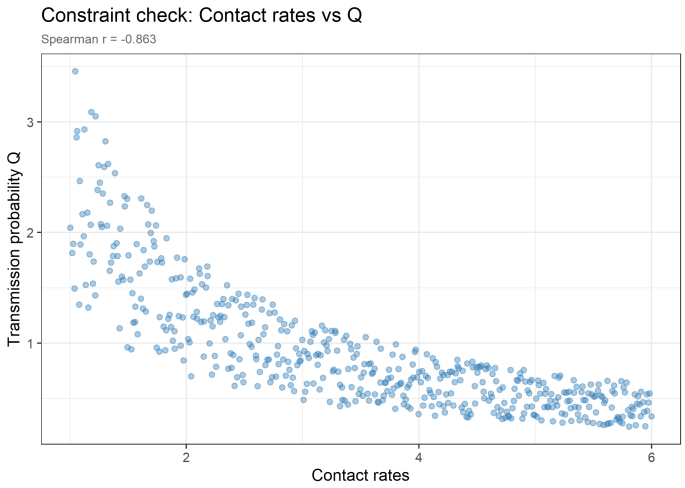


Figure 11: Relationship between contact rates and transmission probability

The pairwise Spearman correlation matrix confirms the near-independence of all seven sampled input parameters (see Figure 16 in Appendix), where all off-diagonal correlations are negligible in magnitude (maximum $|r| = 0.108$).

4 Discussion and Conclusion

4.1 Discussion

The dominant influence of R_0 and k on outbreak dynamics is consistent with the foundational work of Lloyd-Smith et al. (2005) [5], who demonstrated that individual-level variation in infectiousness, captured by the dispersion parameter k, is a critical determinant of epidemic behaviour, particularly for diseases exhibiting superspreading events. The range of k values explores in this study (0.1–1.0) encompasses the estimates reported for SARS-CoV-2 ($k \in [0.1, 0.5]$ [5, 6]), and the finding that low k values produce highly variable, extinction-dominated dynamics aligns with the overdispersion patterns described in these studies. Moreover, the time-varying contact structure implemented in this model draws on empirical evidence from hospital settings [11], adapted to the specific operational schedule of the Vaccinopolis facility. While direct empirical contact data from Vaccinopolis are not yet available, the use of role-specific multipliers across six daily activity periods provides a more realistic representation of contact dynamics than constant-rate models.

4.2 Limitation

Several limitations should be acknowledged in this thesis. The population size of 28 individuals introduces substantial stochastic variability in all outcomes. This leads to high

variance and overlapping confidence intervals across scenarios, meaning the descriptive trends observed cannot be loosely generalized to larger public communities without extensive statistical validation.

Moreover, several core epidemiological parameters are held constant based on static literature values to isolate the effects of contact structures and vaccination. This setup does not account for the natural biological variation across different viral variants or individual host immune health, which could alter the temporal dynamics shown in daily incidence spikes.

Besides, the model applies the vaccine efficacy levels VE_{sus} that takes effect immediately for the vaccinated cohort. It leaves out the real-world complexities of immune dynamics, such as the multi-week time lag required to develop neutralizing antibodies post-injection, the gradual waning of immunity over time, or partial efficacy variations between different vaccine brands.

In addition, the time-block contact multipliers assigned to staff and volunteers (Table 1) are assumption-based, reflecting expected activity patterns rather than empirically measured contact intensities within the Vaccinopolis facility. If empirical contact data become available, these multipliers could be calibrated to facility-specific measurements, potentially improving the realism of the contact structure and enabling validation of the assumed role-specific patterns.

Finally, the primary aim of this thesis is to develop a flexible transmission modelling framework for the Vaccinopolis setting; the specific parameter values serve as a case study to demonstrate the model's capabilities rather than as definitive epidemiological estimates. The current parameterization is framed around COVID-19-like transmission dynamics, so further refinement and extension of SEIR model could be done to assess the disease transmission in different levels exposure to virus or across multiple virus strains. The Latin Hypercube Sampling ranges were chosen broadly enough to encompass both COVID-like ($R_0 \in [1.5, 4.0]$, $k \in [0.1, 0.5]$) and influenza-like (lower R_0, higher k) parameter profiles, but explicit scenario analyses for specific pathogens of interest including respiratory syncytial virus (RSV) or influenza would extend the practical applicability of the model.

4.3 Conclusion

In conclusion, the stochastic individual-based SEIR model that applied in this thesis allows us to capture the essential features of pathogen transmission within the small, closed clinical research facility like Vaccinopolis. We conclude that the severity of outbreaks is influenced by mixing contact structures. Therefore, strict grouping with dedicated staff yields the highest effectiveness in preventing the epidemic transmission, and partial vaccination of volunteers gives further protection, especially useful in settings where cross-contact pathways between groups still exist. Finally, we need to emphasize that the basic reproduction number R_0 and the infectivity heterogeneity parameter k are consistently identified as dom-

inant factors of outbreak dynamics, highlighting the importance of precisely characterizing pathogen transmissibility and superspreading potential for risk assessment. These findings offer a quantitative foundation for guiding staffing plans, vaccination guidelines, and quarantine procedures in the design and conduct of human infection trials at Vaccinopolis.

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A Appendix

A.1 Codes

The full simulation code for this thesis can be found via GitHub at: `https://github.com/tcp7497/Master-Thesis---Chau-`

A.2 Tables

Table 6: Probability of major outbreak by Scenarios and Vaccine efficacy levels

Scenarios	**Placebo**	**VE=30%**	**VE=50%**	**VE=70%**	**VE=90%**
Scenario 1	0.175	0.160	0.155	0.145	0.136
Scenario 2	0.169	0.154	0.142	0.131	0.117
Scenario 3	0.166	0.158	0.148	0.134	0.122
Scenario 4	0.168	0.160	0.150	0.135	0.123

Table 7: Random forest model performance metrics by scenario and outcome

Scenario	**Outcome**	$\mathbf{R^2}$	**OOB** $\mathbf{R^2}$	**OOB MSE**
1: Free mixing	AR_{Total}	0.452	0.452	0.167
	$AR_{Volunteers}$	0.443	0.443	0.172
	AR_{Staff}	0.381	0.381	0.222
2: Staff-bridge	AR_{Total}	0.489	0.489	0.131
	$AR_{Volunteers}$	0.492	0.492	0.129
	AR_{Staff}	0.238	0.238	0.342
3: Strict-cohorting	AR_{Total}	0.412	0.412	0.080
	$AR_{Volunteers}$	0.411	0.411	0.083
	AR_{Staff}	0.221	0.221	0.217
4: Common staff room	AR_{Total}	0.478	0.478	0.091
	$AR_{Volunteers}$	0.464	0.464	0.090
	AR_{Staff}	0.301	0.301	0.285

A.3 Figures

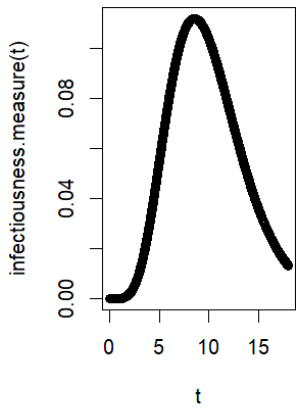


Figure 12: Baseline Gamma-distributed infectiousness profile $\nu(t)$ over 18 days

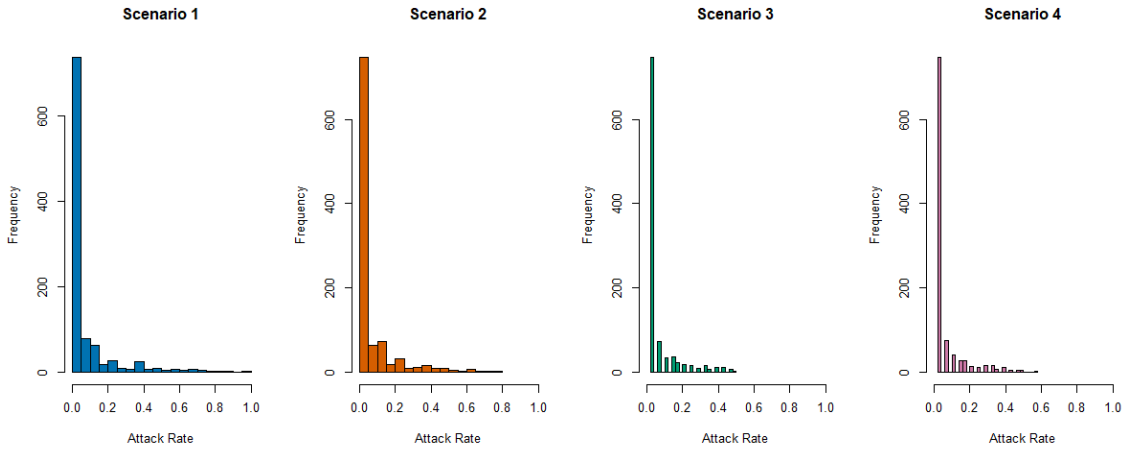


Figure 13: Total attack rates of four scenarios

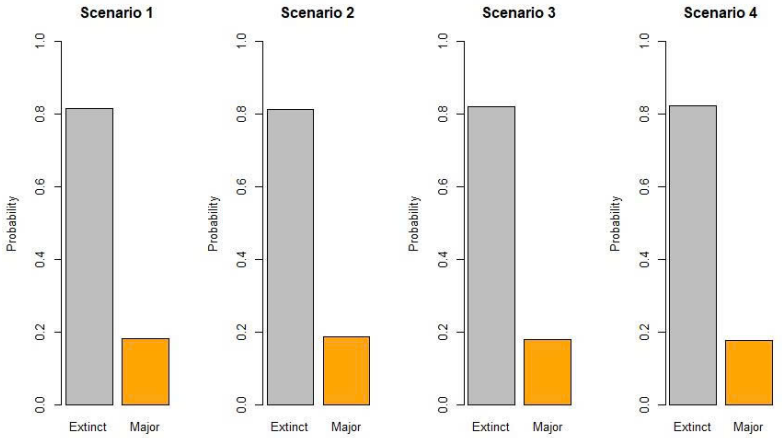


Figure 14: Outbreak probabilities of four scenarios

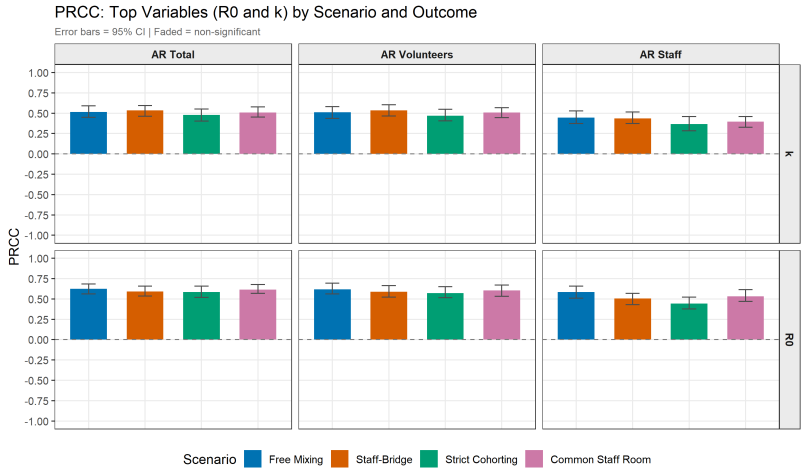


Figure 15: PRCC: Top variables by scenarios and outcomes

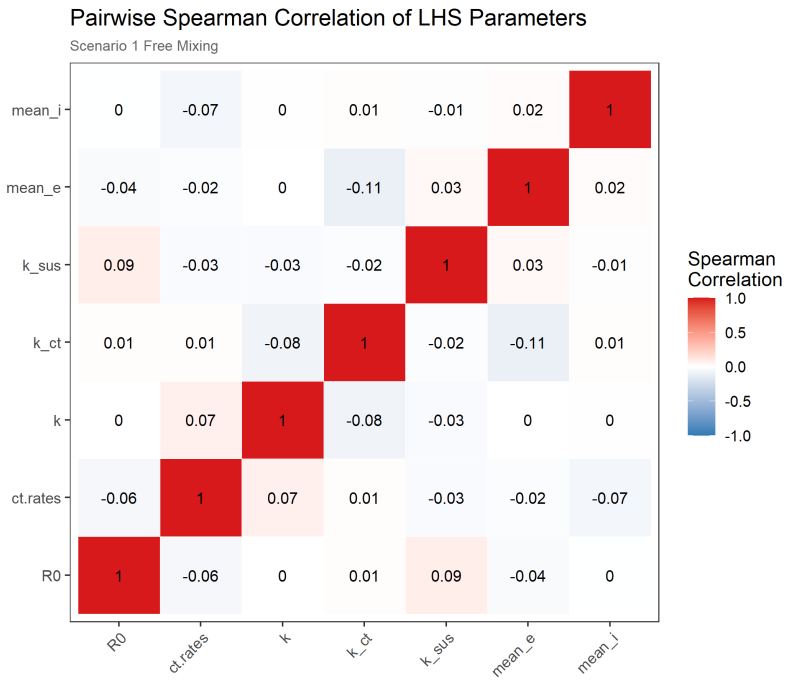


Figure 16: Pairwise Spearman correlation of LHS parameters